Points of intersection

1

a 2 **b** 1 **c** 0 **d** 1

e 0 **f** 2 **g** 1 **h** 1

i 0 **j** 0

2 $(3,4),(-2,9)$

3 Minimum value of $k = 1$

4

a $4x^2 - 5x + 14 = 0$

b -199

c No, because the discriminant is negative which means the quadratic has no solution.

5 5

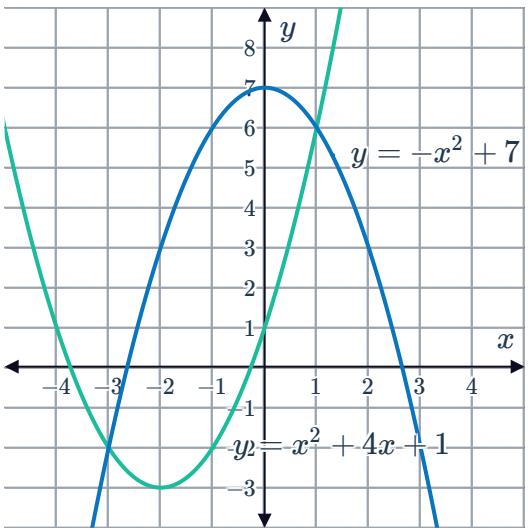
6 $(-5,-67)$

7 $k = 7$

8

a

i

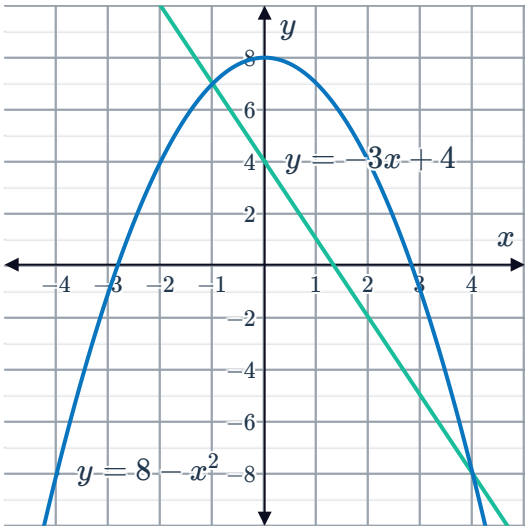


ii $(1,6),(-3,-2)$

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b

i



ii $(-1, 7), (4, -8)$

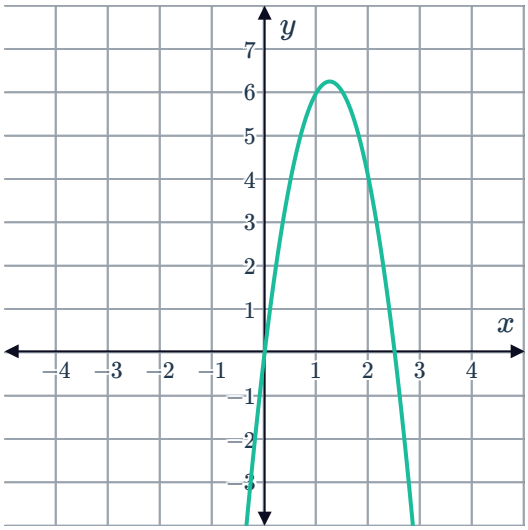
9

a $x = 0, \frac{5}{2}$

b $x = \frac{5}{4}$

c $\left(\frac{5}{4}, \frac{25}{4}\right)$

d



e $c = 6.25$

f $c < \frac{25}{4}$

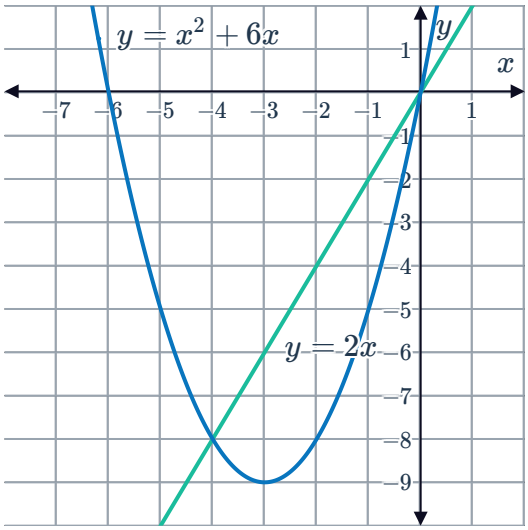
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10

a

x	-4	-3	-2
$y = x^2 + 6x$	-8	-9	-8
$y = 2x$	-8	-6	-4

b



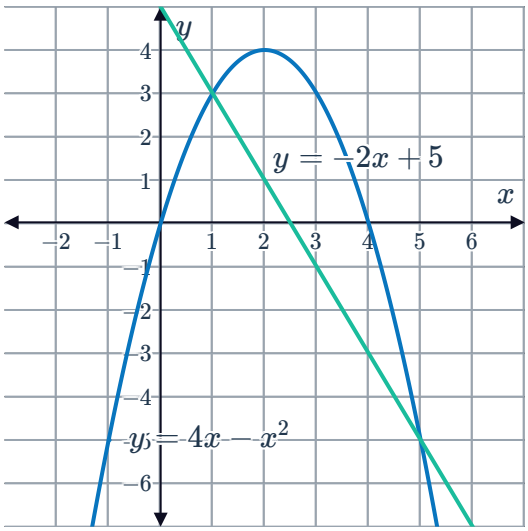
c $(0, 0), (-4, -8)$

11

a

x	1	2	3
$y = 4x - x^2$	3	4	3
$y = -2x + 5$	3	1	-1

b



c $(1, 3), (5, -5)$

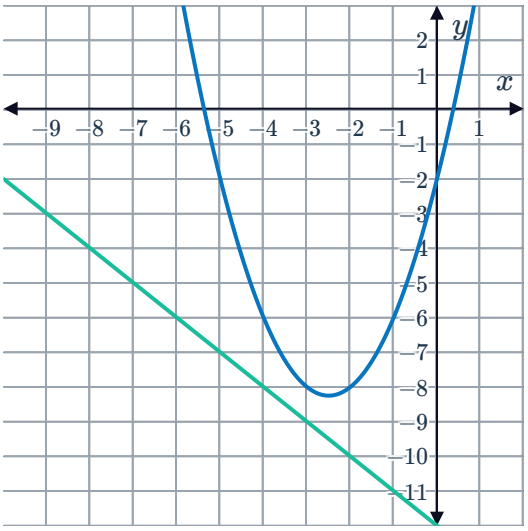
12

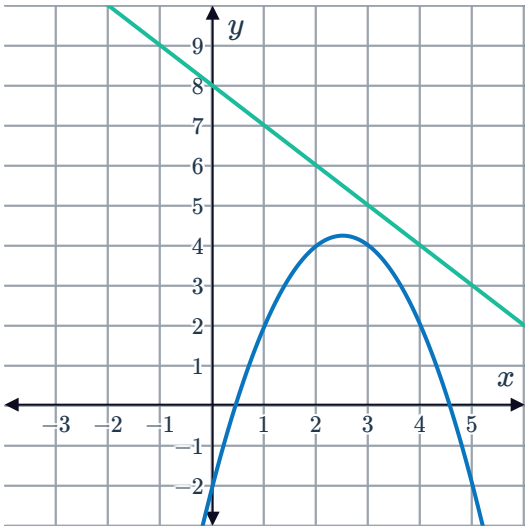
a No

b $(-2, 7)$

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13 Example answers:

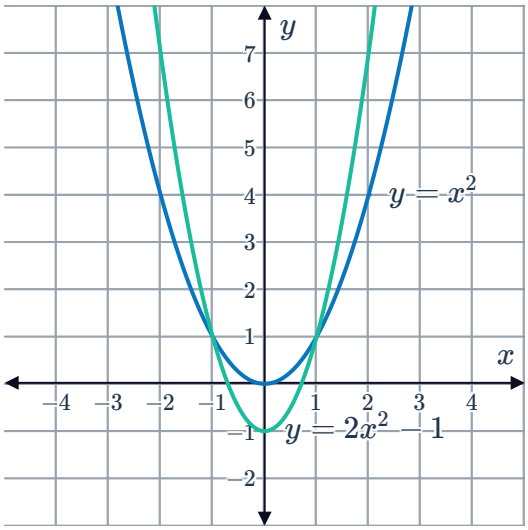




14

a

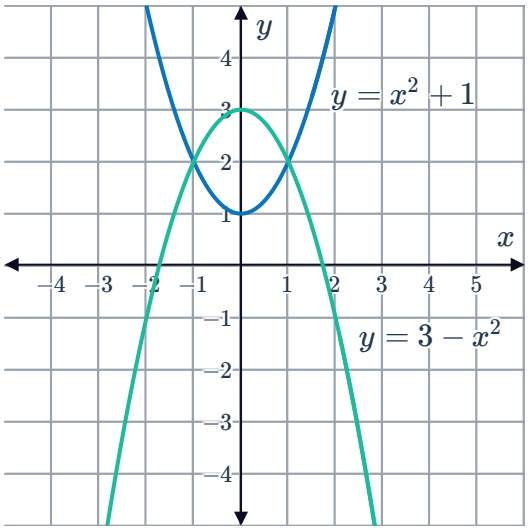
i



ii $(-1, 1), (1, 1)$

b

i



ii $(-1, 2), (1, 2)$

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Quadratic systems of equations

15

a
- i $x = \pm 2$
- ii $(-2, 0), (2, 0)$

b
- i $x = \pm 2$
- ii $(-2, 4), (2, 4)$

c
- i $x = 0$
- ii $(0, 0)$

d
- i $x = 3, -5$
- ii $(3, 25), (-5, 9)$

e
- i $x = 7$
- ii $(7, 8)$

f
- i $x = 5, 4$
- ii $(5, 54), (4, 38)$

g
- i $x = \frac{1}{4}$
- ii $\left(\frac{1}{4}, -\frac{109}{16}\right)$

h
- i $x = \frac{21}{2}$
- ii $\left(\frac{21}{2}, \frac{5}{4}\right)$

i
- i $x = 2$
- ii $(2, 6)$

j
- i $x = \pm\sqrt{10}$
- ii $\left(-\sqrt{10}, -8\right), \left(\sqrt{10}, -8\right)$

16

a $(0, 0), (-7, 49)$ **b** $(3, 18), (-7, 28)$ **c** $(3, -7)$ **d** $(0, 0), (1, 16)$

e $(7, -7), (1, -7)$ **f** $(-4, 42)$ **g** $(2, 2), (-3, -3)$ **h** $(3, 0), \left(\frac{7}{2}, \frac{1}{4}\right)$

i $(2, 1), (-2, 1)$

17

a $x = \frac{1 + \sqrt{17}}{2}, \frac{1 - \sqrt{17}}{2}$

b $\left(\frac{1 + \sqrt{17}}{2}, -4\right), \left(\frac{1 - \sqrt{17}}{2}, -4\right)$

18 $(0.16, -1.15), (-6.16, 5.17)$

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19

a $(-2, 3), (2, -9)$

b $(-4, -12), (-3, -12)$

c $(2, -4), (0, 0)$

d

i $x = 4, -3$ **ii** $(4, 0), (-3, 21)$

e

i $x = -4, -5$ **ii** $(-4, 2), (-5, 5)$

20 $y = -x - \frac{21}{4}$

21

a $x = -4 \pm 2\sqrt{3}$

b

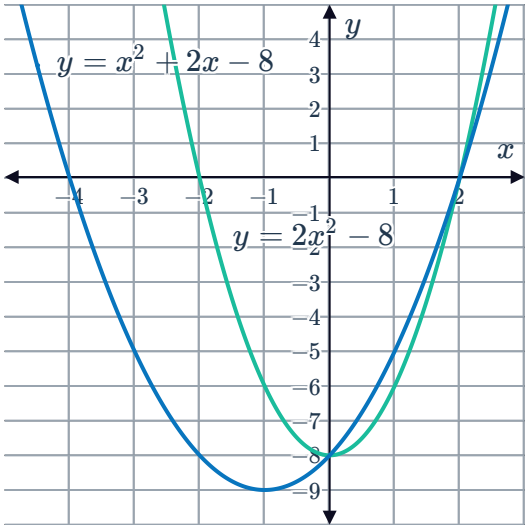
i Yes **ii** Yes **iii** Yes

22

a $c = -8$

b $a = -2$

c



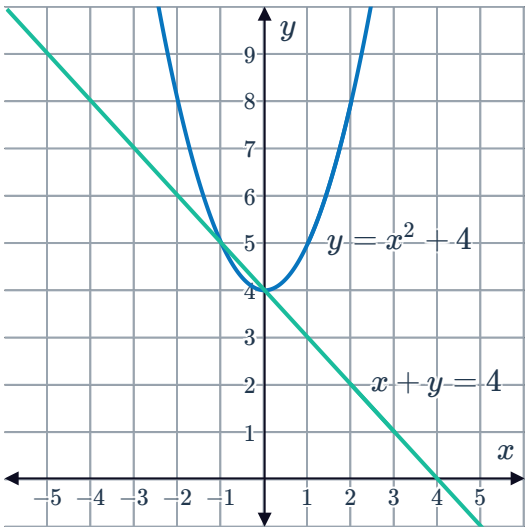
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Quadratic equations from systems

23 $3x^2 - 2x + 9 = 0$

24

a



b 2

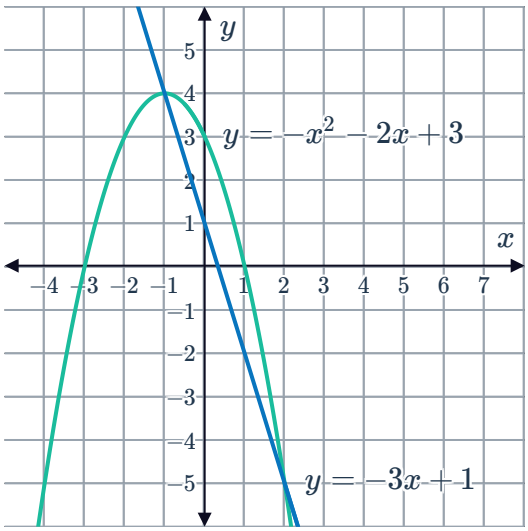
c $(0, 4), (-1, 5)$

25 $y = 2x + 6$

26

a $-3x + 1$

b



c $x = -1, 2$

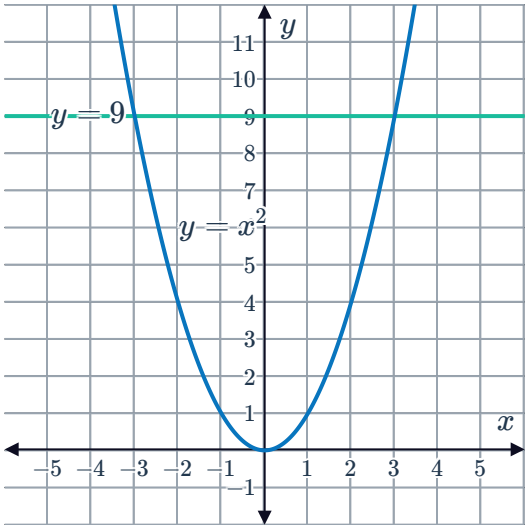
27

a 2 **b** $(0, 0), (-3, 9)$ **c** $x = 0, -3$

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28

a



b $(-3, 9), (3, 9)$

c $x = \pm 3$

Applications

29 $y = 1, \frac{8}{3}$

30

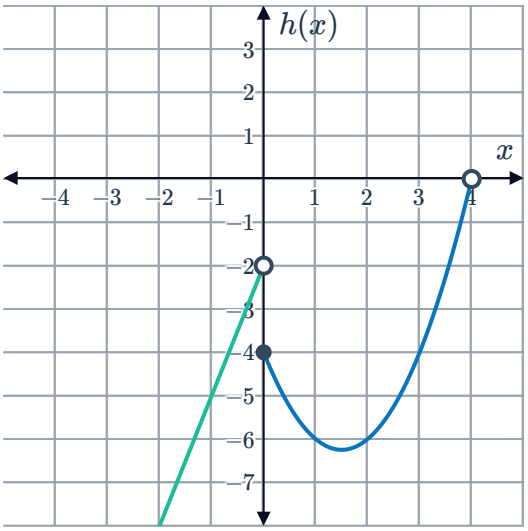
a $x = 0, x = 6$

b $y = 144$

c Example answer:
The equilibrium quantity occurs when 6 thousand units are being produced, which has a corresponding price of 144 dollars per unit.

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